

# Algoritmos y Estructuras de Datos

## CURSADA 2023 - Clases grabadas de años anteriores

|                          |                                                                                                                                                                                                                                                                                                                                                |
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| MÓDULO 1: Árboles y Heap | CLASE 1: Conceptos básicos de Análisis de algoritmos y Repaso de Recursión<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1647373449093">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1647373449093</a>  |
|                          | CLASE 2: Clases e instancias. Tipos de datos - Pasaje de parámetros en JAVA<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1647467861503">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1647467861503</a> |
|                          | CLASE 3: Herencia y clases abstractas en Listas. Tipos genéricos<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1648074024229">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1648074024229</a>            |
|                          | CLASE 4: Árboles Binarios. Definiciones. Recorridos - Árboles de Expresión<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1648581891810">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1648581891810</a>  |
|                          | CLASE 5: Árboles Binarios y de Expresión. Estructura en JAVA y ejemplos<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1648678906933">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1648678906933</a>     |
|                          | CLASE 6: Construcción de Árboles de Expresión – Árboles Generales<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1649187057708">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1649187057708</a>           |
|                          | CLASE 7: Interfaces en JAVA - Objetos Comparable<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1649283484563">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1649283484563</a>                            |
|                          | CLASE 8: Árboles Generales en JAVA - Estructura y ejemplos<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1650493041535">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1650493041535</a>                  |
|                          | CLASE 9: Cola de Prioridades<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1649792059858">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1649792059858</a>                                                |
|                          | CLASE 10: Cola de Prioridades (continuación) y HeapSort<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1650396557216">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1650396557216</a>                     |
|                          | CLASE 11: HEAP en JAVA<br><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1651097706189">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/cd52c3082a873aedccf2982d52b8c3c1e3088541-1651097706189</a>                                                      |

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| <b>MÓDULO 2: TE</b>     | <p>CLASE 12: Cálculo <math>T(n)</math> - Algoritmos iterativos (continuación) y recursivos</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1650999086635">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1650999086635</a></p>                                           |
|                         | <p>CLASE 13: Cálculo <math>T(n)</math> - Algoritmos iterativos (continuación) y recursivos</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1651606154860">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1651606154860</a></p>                                           |
|                         | <p>CLASE 14: Cálculo <math>T(n)</math> - Algoritmos recursivos, Notación Big-Oh, Soluciones iterativas vs. recursivas</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1652210951650">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1652210951650</a></p>                |
|                         | <p>CLASE 15: Cálculo <math>T(n)</math> - Repaso: Explicación de una recurrencia (Ejercicio de las diapos 25 y 26 del video anterior)</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1652307454078">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/b0cc8bdb398afe0b8ab38f6a3cbcb7411c2d0295-1652307454078</a></p> |
| <b>MÓDULO 3: Grafos</b> | <p>CLASE 16: Introducción a Grafos. Definiciones y Terminología</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1652311220985">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1652311220985</a></p>                                                                      |
|                         | <p>CLASE 17: Terminología (cont.) y Recorridos DFS y BFS</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654025459229">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654025459229</a></p>                                                                             |
|                         | <p>CLASE 18: Grafos en JAVA</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654121680202">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654121680202</a></p>                                                                                                          |
|                         | <p>CLASE 19: Orden Topológico de Grafos</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654629832083">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654629832083</a></p>                                                                                              |
|                         | <p>CLASE 20: Caminos Mínimos</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654726465666">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1654726465666</a></p>                                                                                                         |
|                         | <p>CLASE 21: Grafos: caminos mínimos (continuación) - Árbol de expansión mínima: Algoritmos de Prim y Kruskal</p> <p><a href="https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1655331073884">https://bigbluebutton.linti.unlp.edu.ar/playback/presentation/2.3/fcf3d244f9de17c24c5ba56673d1cf5c35f57909-1655331073884</a></p>                        |